lsurvey Tutorial

This tutorial teaches `lsurvey` by building a small project from scratch and then branching into short focused exercises for the commands that do not fit naturally into one workflow.

It assumes you are new to this application. It does not assume you already know how `lsurvey` stores projects, how point IDs work, or how the command line is used inside the TUI.

1. Start The Application

Start a new in-memory project:

```
go run ./cmd/lsurvey
```

Or, after building a binary:

```
./lsurvey
```

When the TUI opens, the screen has four areas:

- `Info` at the top for project name, file path, precision, filter, and status.
- `Points` for stored points.
- `Lines` for stored lines and contour summaries.
- `Command` at the bottom for entering commands.

Useful keys:

- `F1` opens searchable command help.
- `F2` toggles the ASCII map.
- `F3` opens the code styling editor.
- `F4` opens the coordinate conversion workspace.
- `F5` opens a file browser for `.srv` , `.csv` , or `.geojson` files.
- `Tab` completes the current command. When the command input is blank, it cycles the active main pane.
- `Ctrl+n` and `Ctrl+p` cycle completion suggestions.

- `Up` and `Down` browse command history.
- `Alt+s` cycles the point sort field.
- `Alt+d` toggles ascending/descending point sort direction.
- `/` starts a `filter` command.
- Mouse wheel scrolling works in the point list, line list, help, and code styling tables.
- Click the top tabs to switch between Project, Help, Map, Styles, and Convert.
- `Esc` closes help, map, code styling, conversion, or file browsing.
- `Ctrl+C` exits.

Type commands in the bottom command area and press `Enter`.

2. Create Your First Project

Start with a named project:

```
new Lot 42
desc Boundary survey Lot 42
precision 3
units angle=dd.mmss distance=m
```

What these do:

- `new` resets the in-memory project.
- `desc` sets the label shown in the top bar and saved in the project file.
- `precision` changes display formatting only. Stored coordinates are not rounded.
- `units` changes the labels stored with the project.

Save the project:

```
save lot42
```

`lsurvey` appends `.srv` automatically, so this writes `lot42.srv`.

Save the same project under another name:

```
saveas lot42_backup
```

Open a saved project later with or without the extension:

```
open lot42
open lot42.srv
```

3. Add And Edit Points

Points are stored by ID. IDs do not have to be numeric, but numeric IDs are convenient because command completion suggests the next unused number.

Add a few 2D and 3D points:

```
pt add 1 2000 1000 25.500 PEG
pt add 2 2100 1000 25.650 PEG
pt add 3 2100 1080 25.880 PEG
pt add 4 2000 1080 25.700 PEG
pt add 5 2050 1040 TREE
```

The `pt add` forms are:

- `pt add <id> <east> <north> [code]`
- `pt add <id> <east> <north> <elev> [code]`

Edit point fields:

```
pt edit 5 desc="large tree near center"
pt edit 5 code=TREE
pt edit 5 elev=25.740
```

Rename a point:

```
pt rename 5 50
```

List the number of points:

```
pt list
```

Delete a point that is not referenced by a line:

```
pt del 50
```

Re-add it so later examples still work:

```
pt add 50 2050 1040 25.740 TREE
pt edit 50 desc="large tree near center"
```

4. Basic Measurements

With points in place, run a few report-only calculations.

Measure between two points:

```
inverse 1 2
inverse 1 3
```

`inverse` reports azimuth, horizontal distance, easting and northing deltas, and 3D values when both points have elevations.

Measure an angle at a vertex:

```
angle 1 2 3
```

Do bearing math:

```
bearing add 350.0000 20.0000
bearing sub 10.0000 20.0000
```

Do distance math:

```
dist add 12.5 3.125
dist sub 12.5 15
```

5. Create New Points From Calculations

Create a point by direction and distance:

```
rad 1 90.0000 25 as 6 CALC
rad 1 N 45.0000 E 30 vdiff 0.5 as 7 PEG
```

`rad` uses either an azimuth like `90.0000` or a quadrant bearing like `N 45.0000 E`. `vdiff` is optional and only affects elevation when the start point already has one.

Create a 3D point by slope distance and zenith angle:

```
rad3d 1 90.0000 25 90.0000 as 8 SHOT
```

Zenith angle uses `90°` as horizontal. Smaller angles go up; larger angles go down.

Create a midpoint:

```
midpoint 1 3 as 9 MID
```

Create offset points:

```
offset 1 2 5 25 as 10 OFF
offset 1 2 -5 25 as 11 OFF
```

Positive offset is to the left of the direction from the first point to the second point.

6. Store And Manage Lines

Create stored lines:

```
line add B1 1 2 BOUNDARY
line add B2 2 3 BOUNDARY
line add B3 3 4 BOUNDARY
line add B4 4 1 BOUNDARY
line add D1 1 3 EASE
```

Edit a line:

```
line edit D1 code=DIAG desc="lot diagonal"
```

List the number of lines:

```
line list
```

Generate lines automatically from point code:

```
pt add 100 2200 1000 PEG
pt add 101 2250 1030 PEG
pt add 102 2300 1010 PEG
line gen PEG
```

Delete one of those generated lines if you no longer need it:

```
line del L1
```

Point deletion is protected. If you try to delete a point used by a stored line, `lsurvey` rejects it until the referencing line is edited or deleted.

You can also offset from a stored line:

```
offset B1 5 25 as 12 OFF
```

7. Use The TUI Features While You Work

Try filtering the point list:

```
filter PEG
clear filter
```

Sort the points:

```
sort code
sort north desc
```

Open help:

```
help
help rad
help contour gen
```

Press `F1` to open the searchable help browser. Use `/` to filter commands, `Enter` to open the selected command, and `Esc` to return to results or close help. The help browser and command detail pages support arrow keys, page keys, and mouse wheel scrolling.

Open the map with `F2`, the Map tab, or:

```
map
map lines
map contours
map fit
map zoom in
map zoom out
```

Map keys:

- Arrow keys pan.
- `c` toggles stored contour overlays; `~` marks minor and `=` marks index contours.
- `i` toggles point labels between ID and code. Maps start with point IDs.
- `+` or `=` zooms in.
- `-` zooms out.
- `f` fits to points.
- `l` toggles line overlay.
- `F1` opens help from the map.
- `Esc` or `F2` returns to the main view.

Press `F3`, click the Styles tab, or run:

```
style
```

The code styling editor manages style groups, point-code defaults, reusable code libraries, and nominal AutoCAD ACI color previews. Use `g` to add a style group and `c` to add a point-code default from any styling pane. Use `Tab` to move between panes; the group and point-code tables show headers and row ranges, and scroll with arrow keys, page keys, or the mouse wheel.

Press `F4`, click the Convert tab, or run:

```
convert
```

The coordinate conversion workspace stages MGA94/MGA2020 and GDA94/GDA2020 geographic conversions before committing results to the project. Use `s` and `t` for source and target systems, `[` and `]` for MGA source zone, `a` to add a row, `i` to browse for a conversion CSV, `g` to browse for an official NTv2 `.gsb` grid, `r` to calculate, and `c` or `C` to commit one row or all rows. Committed conversion output must be MGA coordinates; geographic-to-MGA batches derive the target zone from longitude and cannot span multiple zones.

8. Intersections And Resection

These commands are easier to understand in a clean mini-scenario. Start a fresh project:

```
new Intersection Demo
pt add A 0 0
pt add B 100 0
pt add C 0 100
pt add D 100 100
```

Line-line intersection:

```
line intersect A D B C as X IP
```

Bearing-bearing intersection:

```
intersect bearing-bearing A 45.0000 B 315.0000 as X1 IP
```

Distance-distance intersection:

```
intersect distance-distance A 100 B 100 choose left as X2 DD
intersect distance-distance A 100 B 100 choose right as X3 DD
```

Bearing-distance intersection:

```
intersect bearing-distance A 45.0000 B 70 choose near as X4 BD
intersect bearing-distance A 45.0000 B 70 choose far as X5 BD
```

Resection uses three known points and bearings observed from the unknown point to those known points:

```
new Resection Demo
pt add 1 290000.504 6147540.603
pt add 2 289477.373 6147660.737
pt add 3 290192.164 6147882.102
resect 1 150.4157 2 246.4213 3 28.1626 as 4 RS
```

9. Traverse Workflow

Start another fresh mini-scenario:

```
new Traverse Demo
pt add 1 5000 5000 100.000
trav start 1
trav leg 90.0000 50 TRV
trav leg 180.0000 40 vdiff -0.2 TRV
trav show
```

Close the traverse onto a known point:

```
pt add 99 5050 4960 99.800
trav close 99
```

Adjust it:

```
trav adjust compass
```

You can also use:

```
trav adjust transit
```

10. Whole-Project Transformations

These commands change stored project coordinates, so work on a copy or a throw away project when learning them.

```
new Transform Demo
pt add 1 1000 1000 10
pt add 2 1100 1000 10
pt add 3 1000 1100 11
line add L1 1 2 TEST
```

Shift every point and stored contour vertex by moving point `1` onto target coordinates:

```
shift 1 east=1005 north=998 elev=10.5
```

Rotate every point and stored contour vertex around a base point:

```
rotate 1 15.3000
rotate 1 -15.3000
```

`rotate` accepts signed compact DMS or signed decimal degrees such as `-15.5d`. It does not accept quadrant bearings.

Convert MGA-style grid coordinates to an anchored local-ground system using a combined scale factor, then reverse that persisted conversion when needed:

```
scale apply 1 csf=0.9996 system=MGA2020_ZONE50
scale reverse
```

Only horizontal coordinates and horizontal contour-distance diagnostics are scaled. Elevations remain unchanged. Export files contain the active coordinates; the TUI status and export message show the label while the scale is applied and remove it after reversal. If the project is shifted or rotated while scaled, the saved scale anchor is transformed with it.

Fit a source-to-target similarity transformation without modifying project data:

```
new Transform Fit Demo
pt add A1 0 0 1
pt add A2 100 0 1
pt add A3 0 100 2
pt add B1 500 200 6
pt add B2 500 80 6
pt add B3 620 200 7
transform fit A1 B1 A2 B2 A3 B3
```

`transform fit` reports:

- `dx` and `dy`
- optional `dz`
- signed angle
- floating-point scale
- horizontal RMS residual
- vertical RMS residual when complete 3D pairs exist

11. Generate And Manage Contours

Contours require at least three elevated, non-collinear points. This section uses a small scratch dataset created from scratch.

```
new Terrain Demo
pt add 1 0 0 100 BRK
pt add 2 50 0 101 BRK
pt add 3 100 0 102 BRK
pt add 4 0 50 100.5
pt add 5 50 50 102
pt add 6 100 50 103
pt add 7 0 100 101
pt add 8 50 100 103
pt add 9 100 100 104
line add R1 1 2 RIDGE
line add R2 2 3 RIDGE
```

Generate contours using all lines as breaklines:

```
contour gen C1 1
```

Generate contours from points only:

```
contour gen C2 0.5 breaklines=none
```

Generate contours using selected breaklines only:

```
contour gen C3 0.5 base=100 index=5 breaklines=ids:R1,R2
```

Clip contours to closed linework selected by code, with optional exclusions:

```
contour gen C4 0.5 boundary=codes:SITE exclude=codes:POND
contour regen C4
```

Request quality warnings for long TIN edges and optional presentation smoothing:

```
contour gen C5 0.5 boundary=codes:SITE maxedge=30 smooth=1
contour info C5
```

Inspect and manage contour sets:

```
contour list
contour info C1
contour del C2
```

Important contour rules:

- Points without elevations are ignored.
- Breakline endpoint points must have elevations.
- Selected breaklines must not cross except at shared endpoints.
- `boundary=codes:` and `exclude=codes:` use closed stored-line rings; their points may be 2D because they clip contours horizontally.
- Relevant point or line changes mark stored contour sets stale; run `contour regen <id>` to rebuild them from the saved generation settings.
- Contour generation records warnings for duplicate elevated positions, long TIN edges, and unconstrained hull contact. Omitting `maxedge=` uses an automatic threshold based on point spacing.

- `smooth=1` through `smooth=3` stores smoothed output while retaining raw contour geometry; DXF uses the smoothed output.

12. Import And Export

Save the current project before exporting:

```
save terrain_demo
```

CSV import and export:

```
export csv terrain_points
import csv terrain_points
```

CSV files must use this exact header:

```
id,easting,northing,elevation,code,description
```

Export GeoJSON:

```
export geojson terrain
```

Import GeoJSON:

```
import geojson terrain
```

Export DXF:

```
export dxf terrain
```

DXF includes points, point labels, stored lines, and stored contours. Contour labels are placed along readable contour paths and omitted when a path is too short or would overlap an existing contour label.

Export LandXML when the project distance unit is metric:

```
export landxml terrain
```

Export polygon area and boundary schedules:

```
export boundarycsv terrain_boundaries
export boundarycsv lot1_boundary polygon=LOT1
```

Reusable code-style libraries use `.codes.json` :

```
export codes field_styles
import codes field_styles
```

13. Batch Export Without Opening The TUI

Once a project has been saved, you can export it from the shell:

```
go run ./cmd/lsurvey export csv ./terrain_demo.srv ./terrain_points
go run ./cmd/lsurvey export dxf ./terrain_demo.srv ./terrain
go run ./cmd/lsurvey export geojson ./terrain_demo.srv ./terrain
```

These commands append the output extension when missing.

14. Completion, History, And Practical Workflow

As you work:

- Use `Tab` to advance through a completion one argument at a time.
- With an empty command input, use `Tab` and `Shift+Tab` to move focus between the command, point list, and line list panes.
- Use `Ctrl+n` and `Ctrl+p` to cycle suggestions.
- Use `Up` and `Down` to revisit earlier commands.
- Use `Alt+s` and `Alt+d` to change point sorting without typing a command.
- Use `F1` through `F4` for Help, Map, Styles, and Convert, or click the top tabs. Use `F5` to browse for project, CSV, or GeoJSON files.
- Use the mouse wheel to scroll long point lists, line lists, help pages, and code styling tables.
- Use `help <command>` whenever you forget the exact syntax.

A practical day-to-day flow often looks like this:

```
new Job Name
desc Boundary survey
precision 3
pt add ...
line add ...
inverse ...
offset ...
map lines
map contours
save job_name
export dxf job_name
export csv job_name_points
```

15. Full Command Checklist

This section is a quick index so you can confirm what the application supports.

Project:

- `new`
- `open`
- `save`
- `saveas`
- `desc`
- `precision`
- `units`
- `undo`
- `redo`
- `history`
- `history info`
- `info`
- `quit`
- `exit`

Points:

- `pt add`
- `pt edit`
- `pt del`
- `pt rename`

- `pt list`

Calculations:

- `inverse`
- `angle`
- `close`
- `bearing add`
- `bearing sub`
- `dist add`
- `dist sub`
- `rad`
- `rad3d`
- `midpoint`
- `offset`
- `shift`
- `rotate`
- `scale apply`
- `scale reverse`
- `transform fit`

Intersections:

- `line intersect`
- `intersect bearing-bearing`
- `intersect bearing-distance`
- `intersect distance-distance`
- `resect`

Traverse:

- `trav start`
- `trav leg`
- `trav show`
- `trav close`
- `trav adjust`

Contours:

- `contour gen`
- `contour regen`

- `contour list`
- `contour info`
- `contour del`

Lines and stored geometry:

- `line add`
- `line gen`
- `line edit`
- `line del`
- `line list`
- `polyline add`
- `polyline edit`
- `polyline del`
- `polyline list`
- `polyline info`
- `polygon add`
- `polygon edit`
- `polygon del`
- `polygon list`
- `polygon info`
- `polygon report`

Import and export:

- `import csv`
- `export csv`
- `import geojson`
- `export geojson`
- `export dxf`
- `export landxml`
- `export boundarycsv`
- `import codes`
- `export codes`

Styling and conversion:

- `group add`
- `group edit`
- `group del`

- `group list`
- `code style set`
- `code style del`
- `code style list`
- `style`
- `convert`

TUI helpers:

- `filter`
- `clear filter`
- `sort`
- `map`
- `map lines`
- `map contours`
- `map fit`
- `map zoom in`
- `map zoom out`
- `help`
- `help <command>`
- `F1` help
- `F2` map
- `F3` styles
- `F4` coordinate conversion
- `F5` file browser

If you forget the exact shape of a command, use `help` inside the app. That is the fastest way to confirm the current syntax.